

**LEGALVISION: A KNOWLEDGE-DRIVEN AI FRAMEWORK
FOR LEGAL CONTRACT UNDERSTANDING AND TRUST
ASSESSMENT**

Project ID: 25-26J-127

Project Proposal Report

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Specialized in Data Science

Department of Information Technology
Sri Lanka Institute of Information Technology
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Declaration

Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

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.....2025/8/28.....

Date

ABSTRACT

Property law in Sri Lanka involves complex relationships among statutes, deeds, case precedents, and ownership records, often making legal analysis time-consuming and prone to oversight. Existing digital systems provide basic search and retrieval but lack the semantic connectivity, dynamic updates, and structured reasoning support required for advanced legal decision-making. These limitations hinder the creation of reliable, context-aware tools for legal professionals, registrars, and policymakers.

This proposal introduces a dynamic, auto-updating legal knowledge graph designed specifically for Sri Lankan property law. The system integrates natural language processing (NLP), entity and relation extraction, and graph-based storage to map relationships among legal statutes, property documents, and case precedents. Unlike traditional static systems, this knowledge graph continuously incorporates updates from gazettes, registries, and judgments, ensuring accurate and current legal insights. It also supports query-based retrieval and graph visualizations, enabling intuitive exploration of complex legal relationships.

Through innovations in data structuring, semantic linking, and dynamic updating, this component advances LegalVision’s capabilities beyond document-level analysis toward deep, interconnected legal reasoning. The outcome is a property law knowledge base that enhances transparency, supports explainable AI workflows, and empowers stakeholders with accurate, traceable, and context-rich legal information.

Subject Descriptors (ACM CCS)

- Computing methodologies → Artificial intelligence → Knowledge representation and reasoning → Knowledge graphs
- Applied computing → Law, social and behavioral sciences → Legal informatics

Keywords

Property law, Knowledge graph, Dynamic updates, Legal data integration, NLP, Semantic reasoning

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List of Abbreviations

Table 1: List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
DB	Database
DL	Deep Learning
FAISS	Facebook AI Similarity Search
FastAPI	Fast Application Programming Interface (Python framework)
GDPR	General Data Protection Regulation
GPU	Graphics Processing Unit
HTTP/HTTPS	Hypertext Transfer Protocol / Secure Hypertext Transfer Protocol
JSON	JavaScript Object Notation
JWT	JSON Web Token
KG	Knowledge Graph
LLM	Large Language Model
LoRA	Low-Rank Adaptation
NER	Named Entity Recognition
NLP	Natural Language Processing
OCR	Optical Character Recognition
OIDC	OpenID Connect
PEFT	Parameter-Efficient Fine-Tuning
QoS	Quality of Service
RAG	Retrieval-Augmented Generation
RAD	Rapid Application Development
REST	Representational State Transfer
S3	Simple Storage Service (Object Storage)
SDLC	Software Development Life Cycle
SPARQL	
SSL/TLS	Secure Sockets Layer / Transport Layer Security
UI	User Interface
UX	User Experience
VM	Virtual Machine
NoSQL	Non-Relational Structured Query Language
Neo4j	Graph Database used for entity-relationship modeling

LoE	Level of Effort
CoT	Chain of Thought (prompt reasoning method)
PRINCE2	Projects in Controlled Environments, Version 2 (Project Management Methodology)
React	JavaScript library for building front-end interfaces
OCR	Optical Character Recognition
MongoDB	NoSQL Database for document storage
S3/MinIO	Object Storage for documents and OCR outputs
PDF	Portable Document Format
RDF/OWL	Resource Description Framework / Web Ontology Language
ELI	European Legislation Identifier
QA	Quality Assurance
JSONL	JSON Lines (for structured datasets)
CSV	Comma-Separated Values (data file format)
I/O	Input/Output

1. INTRODUCTION

1.1 Background & Literature Survey

1.1.1 Research Background

Property law in Sri Lanka is complex, integrating Roman-Dutch principles, English Common Law, and statutory frameworks unique to the island. This layered structure governs the interpretation and execution of deeds, title documents, leases, and partition agreements [1]. Effective decision-making in this domain requires access to accurate, connected, and up-to-date legal information—including statutes, gazettes, judgments, and ownership records. However, much of this data is dispersed across manual registries, static legal databases, and fragmented reporting systems, making integrated legal reasoning a significant challenge [2] [3].

A legal knowledge graph offers a solution by representing legal entities—such as properties, owners, statutes, and judgments—and the relationships between them in a machine-readable, interconnected network. In global contexts, such as the U.S. and EU, knowledge graphs have been successfully implemented to improve legal research, automate compliance checks, and support reasoning tasks [4] [5]. These systems enable dynamic querying, semantic search, and relationship analysis, turning unstructured legal texts into structured, actionable insights.

For Sri Lanka, the need for such a dynamic, context-aware knowledge system is pressing. Issues like overlapping claims, incomplete registration, and inconsistent updates in the deeds-based system often result in disputes and prolonged litigation [6]. Efforts like the Title Registration Act of 1998 aimed to modernize property registration, but adoption has been slow and integration with digital systems remains limited [7]. At the same time, open-access resources such as the Sri Lanka Gazette, LankaLaw, and Law & Society Trust archives provide valuable but unlinked datasets that, if integrated, could power more advanced legal analytics and reasoning systems.

Globally, advanced Natural Language Processing (NLP) and graph-based systems have demonstrated the ability to extract entities, identify relationships, and populate knowledge bases dynamically [8] [9]. Combining named entity recognition (NER) with relationship extraction can transform large volumes of unstructured legal text—statutes, case law, and property records—into structured, queryable formats. These graphs can then support advanced workflows such as statute-aware document review, automated compliance checks, and enhanced legal reasoning for AI systems.

Despite these advancements, Sri Lanka lacks a centralized, AI-driven legal knowledge graph tailored to property law. Existing tools are either static repositories or paywalled databases that do not offer real-time updates, semantic linking, or interoperability with reasoning engines. Developing a dynamic legal knowledge graph specific to Sri Lankan property law would address these gaps, enabling transparent, traceable, and contextually accurate legal analysis. Such a system would not only support legal practitioners but also underpin explainable AI modules like LegalVision’s reasoning engine, creating a robust foundation for trustworthy, AI-assisted legal decision-making in the property domain.

1.2 Research Gap

Despite rapid progress in knowledge-graph (KG) methods for law, there is no Sri Lanka-specific, end-to-end legal knowledge graph for property law that integrates statutes, gazettes, case law, cadastral/survey data, and registry records into a machine-readable, continuously updated platform. Existing resources are fragmented across institutions, published largely as static PDFs, and lack common identifiers and interoperable metadata—creating barriers for retrieval, compliance, analytics, and downstream explainable AI. This section details the gap.

1.2.1 Fragmented and siloed legal-land administration data

Land administration in Sri Lanka is institutionally and functionally fragmented, with responsibilities spread across many agencies; this impedes integration and consistent access to authoritative records (e.g., deeds, plans, and notifications). The World Bank’s assessment notes fragmentation and geographic incompleteness in land administration, directly constraining governance and service delivery. Over 50 institutions share land-related mandates, underscoring the lack of a single, linked data backbone [3] [2]

1.2.2 Static, PDF-first legal publishing and limited machine-readability

Core sources—such as the Registration of Documents Ordinance and subsequent amendments—are commonly disseminated as PDFs via the Government Printer and other portals. While authoritative, these formats do not expose structured identifiers, citations, or machine-actionable relationships needed for a KG (e.g., section-to-section links, cross-references to cases). This hinders automated parsing, entity linking, and continuous updates. [10] [1]

1.2.3 Lack of common legal identifiers and interoperability standards

Modern legal KGs in the EU have converged on metadata/identifier standards—for example, ELI (European Legislation Identifier)—to enable cross-system linking, versioning, and provenance at scale. Sri Lanka does not yet publish legislation with ELI-style URIs or RDF/JSON-LD metadata, limiting interoperability and reuse. [11] [12].

1.2.4 Missing domain ontologies aligned to Sri Lankan property law

Global surveys describe robust KG foundations (ontologies, embeddings, quality management), but these are domain-agnostic or foreign-jurisdiction-centric. There is no published ontology that captures Sri Lanka’s property instruments (e.g., deed/encumbrance types), statutory regimes (e.g., Title Registration Act, Land Development Ordinance), and procedural artifacts (e.g., partition actions) with local semantics and constraints. [13] [5] [14].

1.2.5 Limited integration of cadastral/survey information

Property reasoning requires links between legal rights and spatial parcels. Although the Survey Act No. 17 of 2002 empowers the Survey Department to regulate cadastral and geodetic standards, there is no publicly documented, national linked data layer that connects parcels/plans with legal sources (statutes, gazettes, judgments) for KG ingestion. [15].

1.2.6 Absence of a dynamic, Sri Lanka-specific legal KG

EU efforts (e.g., Lynx Legal Knowledge Graph) demonstrate feasibility for multi-jurisdiction KGs, but there is no comparable, open, update-aware KG for Sri Lankan property law that: (i) harvests gazettes and acts as they are issued, (ii) models amendments/versions, (iii) links to case law and registry events, and (iv) exposes query APIs for downstream reasoning engines. [16] [17] [18].

There is no operational, open, and Sri Lanka-tailored property-law KG that unifies statutes, gazettes, case law, cadastral data, and registry records under shared identifiers/ontologies, supports extraction, and provides provenance-rich, API-accessible knowledge for explainable analytics. Building such a KG—aligned to local statutes (e.g., Registration of Documents, Title Registration, Land Development Ordinance) and spatial frameworks (Survey Act)—would close a critical infrastructural gap and enable higher-level applications (e.g., compliance checks, risk flags, and legal reasoning modules) for LegalVision.

Table 2: Absence of a dynamic, Sri Lanka-specific legal KG

System / Feature	Integrated Data	Machine-Readable	Ontology Support	Dynamic KG Access
Current Registries / Static Portals	X	X	X	X
Case Law Databases (Paywalled)	X	Partial	X	X
Gazette Archives (PDF)	X	X	X	X
Cadastral Systems	Partial	X	X	X
International Legal KGs (EU Lynx, etc.)	✓	✓	✓	✓ (non-local)
Proposed LegalVision KG	✓	✓	✓	✓ (localized, real-time)

1.3 Research Problem

1.3.1 Problem Discussion

Sri Lanka’s legal data for property law—spanning statutes, gazettes, case law, and cadastral records—is dispersed, non-standardized, and often locked in static formats such as PDFs or scanned documents. This fragmentation prevents machine-readable integration, real-time updates, and semantic linking of critical legal information. Globally, legal knowledge graphs have been shown to improve reasoning, compliance checks, and advanced analytics [4] [5], yet no such system exists for the Sri Lankan context. Without a dynamic, and domain-specific legal knowledge graph, AI models lack the structured data required for accurate reasoning and explainability. Legal practitioners are forced to rely on manual cross-referencing of statutes and case law, reducing efficiency and increasing the risk of error in property transactions, dispute resolution, and policy planning [2] [3].

1.3.2 Research Questions

1. How can NLP and entity-relation extraction techniques be applied to build a dynamic, machine-readable knowledge graph of Sri Lankan property law?
2. What ontology and schema designs best represent Sri Lankan statutes, precedents, and property entities for interoperability and reasoning?
3. How can real-time updates from gazettes, registries, and judgments be integrated to maintain a continuously relevant and reliable knowledge base?

1.3.3 Problem Statement

There is no dynamic, structured, and domain-specific knowledge graph that integrates Sri Lankan property law statutes, case law, and cadastral data to support AI-driven reasoning and explainability.

2. OBJECTIVES

2.1 Main Objective

To design and implement a dynamic, and domain-specific legal knowledge graph (KG) for Sri Lankan property law that integrates statutes, gazettes, case law, cadastral data, and registry records into a machine-readable, continuously updated, and queryable platform, enabling interoperability, semantic reasoning, and explainability for AI-driven legal applications such as LegalVision's reasoning module.

2.2 Specific Objectives

1. Develop a domain-specific ontology and schema
 - Create a formal ontology to represent entities and relationships in Sri Lankan property law, such as property parcels, ownership records, statutory provisions, precedents, and administrative actions.
 - Align the ontology with international standards (e.g., ELI, RDF/OWL) to enable future interoperability.
2. Build a comprehensive legal data pipeline
 - Collect and preprocess primary legal sources including statutes, amendments, gazettes, case law, cadastral data, and registry records.
 - Normalize unstructured formats (e.g., PDFs, scanned deeds) into structured, machine-readable data suitable for KG ingestion.
3. Design and populate the knowledge graph
 - Use graph database technologies such as Neo4j or RDF-based stores to store entities, relationships, and metadata.
 - Incorporate version control and provenance to track amendments, judgments, and ownership changes over time.
4. Enable dynamic updates and continuous integration

- Develop automated pipelines to harvest and update legal data from sources such as the Government Gazette, Registry Department, and court databases.
 - Ensure real-time or scheduled refreshes to maintain up-to-date, reliable information.
5. Integrate APIs for reasoning and analytics
- Build RESTful and GraphQL APIs to make the KG accessible to LegalVision’s reasoning module and other analytics platforms.
 - Support complex queries like:
 - “Retrieve all cases referencing the Registration of Documents Ordinance after 2015.”
 - “List all parcels affected by state acquisition notices in Colombo District.”
6. Validate accuracy and usability
- Conduct evaluations with legal experts to ensure the correctness, completeness, and usability of the KG.
 - Use benchmark queries and test cases to measure performance, accuracy, and latency.
7. Ensure ethical and legal compliance
- Incorporate data privacy and governance measures to comply with the Personal Data Protection Act and professional legal standards.
 - Document all processes for auditability and transparency.

2.3 Outcome of the Objectives

By achieving these objectives, the project will deliver a scalable and trustworthy knowledge infrastructure that:

- Serves as the core data backbone for LegalVision's reasoning module and other property law applications.
- Provides queryable, linked, and explainable insights for legal practitioners, policymakers, and AI researchers.
- Creates a foundation for future innovations in legal analytics, compliance automation, and risk management in Sri Lanka's property sector.

3. METHODOLOGY

3.1 Development Methodology

To support experimentation, modular implementation, and continuous improvements aligned with AI-driven legal reasoning research, the Agile development methodology has been adopted for this project. Agile's sprint-based model provides flexibility to respond to evolving design decisions, integration requirements, and expert feedback from legal practitioners. Each sprint will focus on a specific task—such as data preprocessing, model fine-tuning, API integration, or front-end visualization—allowing for incremental progress, validation, and course correction where needed.

This methodology is preferred over traditional models such as Waterfall due to its adaptability and suitability for research-oriented projects involving interdisciplinary collaboration, technical experimentation, and iterative refinements. The Agile process empowers a feedback-driven and responsive development cycle, ensuring that the reasoning module remains aligned with research goals, technical feasibility, and the practical expectations of legal professionals.

Development Methodologies Comparison

Table 3: Development Methodologies Comparison

Methodology	Description
Waterfall Model	A linear, phase-based approach where each stage must be completed before moving to the next. While structured and easy to manage, it lacks flexibility to accommodate new findings or adjustments during ongoing research. Unsuitable for AI projects that require continuous experimentation and refinement.
RAD (Rapid Application Development)	Focuses on rapid prototyping and fast feedback cycles. However, it is highly resource-intensive and demands consistent user involvement,

	making it less practical for research projects with limited stakeholder availability.
Prototyping Model	Emphasizes building early system versions for stakeholder evaluation and feedback. While beneficial for refining user-facing features, it may not fully align with back-end-heavy, model-centric research that prioritizes performance and reasoning accuracy.
Agile Model	A flexible and iterative approach ideal for research requiring incremental development, model testing, and prompt-driven refinement. Encourages continuous feedback and rapid adaptation to emerging insights. Suitable for AI projects where stages like data preprocessing, model fine-tuning, and evaluation evolve dynamically.

Given the experimental nature of this project and the need for continuous validation across data, reasoning models, and evaluation outputs, the Agile model was selected. Its support for iterative development, modular design, and stakeholder-driven feedback makes it the most suitable approach for building and refining the explainable reasoning module efficiently while maintaining alignment with the project objectives and integration requirements with the knowledge graph.

3.2 Project Management Methodology

For effective planning, execution, and tracking of this research-driven project, the Agile PRINCE2 methodology was adopted. This hybrid model combines the structured governance and documentation rigor of PRINCE2 with the iterative flexibility of Agile, making it highly suitable for an academic research project that involves evolving technical goals, continuous experimentation, and interdisciplinary collaboration. Agile PRINCE2 supports stage-based planning aligned with academic milestones while enabling incremental delivery and frequent validation with supervisors and legal experts. This ensures that the project maintains a balance between academic discipline, risk control, and adaptability to accommodate emerging insights during development and testing.

Justification for Agile PRINCE2

Table 4: Justification for Agile PRINCE2

Methodology	Description	Suitability
PRINCE2	A structured, process-driven methodology focusing on documentation, risk control, and milestone-based planning.	Provides strong governance but lacks the adaptability needed for experimental AI research.
Agile PM	Emphasizes rapid iteration, flexibility, and stakeholder feedback, but with less formal structure.	Highly adaptive but can struggle with academic milestone management and detailed documentation requirements.
Agile PRINCE2 (Selected)	A hybrid approach combining PRINCE2's structure with Agile's responsiveness.	Offers formal project controls while supporting iterative experimentation and continuous improvement, making it ideal for this project.

3.3 Scope

The project scope has been clearly defined to ensure focus, feasibility, and alignment with the research objectives:

3.3.1 In-Scope Activities

- Data collection and preprocessing of Sri Lanka-specific legal texts, including statutes, precedents, and administrative documents.
- Fine-tuning large language models for reasoning and explainability in property law contexts.
- Development of FastAPI-based backend services and React.js front-end interfaces for querying and visualization.
- Integration with the knowledge graph module to enhance reasoning accuracy and traceability.
- Evaluation of reasoning outputs through quantitative metrics and expert reviews.

3.3.2 Out-of-Scope Activities

- Production-grade deployment or commercial rollout of the system.
- Use of sensitive or confidential legal records due to ethical and data privacy constraints.
- Full automation of legal decision-making; human oversight will remain a core requirement.

3.3.3 Milestone Planning and Schedule

The project will be structured into stages and sprints, each aligned with academic and technical deliverables:

Table 5: Milestone Planning and Schedule

Deliverable / Milestone	Planned Date
Topic exploration and brainstorming	July 1 – July 10, 2025
Project topic finalized	July 15, 2025
Topic Assessment Form (TAF) submission	July 20, 2025
Project proposal finalized	August 25, 2025
Literature review drafted and approved	September 15, 2025
System architecture design and tool selection	October 10, 2025
Data collection and preprocessing completed	October 31, 2025
Initial LLM fine-tuning and reasoning module prototype	November 25, 2025
Integration with knowledge graph and front-end UI	December 15, 2025
Testing, debugging, and evaluation completed	January 10, 2026
Make Improvements	January 20, 2026
Final report writing and documentation	January 31, 2026
Final submission and presentation	February 5, 2026

A detailed Gantt chart outlining these milestones is be provided at [Appendix-III](#).

3.3.4 Deliverables

- Clean and structured datasets of Sri Lankan property law documents.
- A functional FastAPI-based reasoning API capable of processing queries and generating transparent reasoning chains.
- An interactive React.js interface for visualizing reasoning steps and legal logic graphs.
- Evaluation reports detailing accuracy, explainability, and expert validation outcomes.
- Comprehensive documentation of the system design, architecture, and evaluation methodologies.

3.4 Implementation Strategy – Knowledge Graph Component

The implementation of the Legal Knowledge Graph (LKG) for Sri Lankan property law will be executed in a multi-phase, modular approach. This strategy ensures that the graph is accurate, scalable, and continuously updatable, providing a robust foundation for advanced reasoning and explainability in LegalVision.

3.4.1 Phase 1 – Data Collection and Preprocessing

A comprehensive, high-quality, and machine-readable dataset is critical to build a reliable LKG. Data source details are added to [Appendix-I](#).

Data Sources

- Statutory Laws: Key ordinances and acts including the Registration of Documents Ordinance, Title Registration Act, Land Development Ordinance, and amendments.
- Case Law: Property-related judgments from the Supreme Court and Court of Appeal archives.
- Gazette Notifications: Acquisition notices, government grants, and regulatory circulars from the Government Printer.

- Cadastral and Survey Data: Parcel-level data from the Survey Department.
- Secondary References: Legal commentary, law journals, and property law textbooks to enrich relationships.

Preprocessing Steps

- OCR and Text Extraction: Converting scanned and PDF legal documents into structured digital text.
- Cleaning and Normalization: Standardizing legal terms, removing duplicates, and segmenting into logical units like sections, subsections, and clauses.
- Entity Tagging: Identifying key entities such as statutes, parties, locations, property types, and case references.

3.4.2 Phase 2 – Ontology and Schema Design

A formal ontology will define the structure and semantics of the LKG:

Core Elements

- Entities: Statutes, parcels, judgments, parties, transactions, and administrative actions.
- Relationships:
 - “is-governed-by” (parcel → statute)
 - “is-cited-in” (case → statute)
 - “amends” (amendment → original statute)
 - “affects” (gazette → parcel)
- Attributes: Dates, jurisdiction, references, and confidence scores for extracted relationships.

Standards

- Ontology design will follow RDF/OWL frameworks and support ELI (European Legislation Identifier)-style identifiers to ensure global interoperability and long-term usability.

3.4.3 Phase 3 – Entity and Relationship Extraction

Advanced Natural Language Processing (NLP) will be applied to populate the KG accurately:

- Named Entity Recognition (NER): Using domain-tuned models (e.g., LegalBERT, spaCy, or HuggingFace transformers) to detect legal entities and references.
- Relation Extraction: Applying deep learning pipelines to detect and label relationships, such as parcel ownership links or statutory amendments.
- Disambiguation and Linking: Matching entities across multiple data sources to avoid duplicates and ensure accurate connections between statutes, cases, and cadastral data.

3.4.4 Phase 4 – Knowledge Graph Construction

The KG will be built on a scalable and queryable platform:

- Technology Stack:
 - Neo4j or GraphDB for graph storage and traversal.
 - MongoDB for document and metadata storage.
- Data Ingestion Pipelines: Automated scripts to populate and update the graph with new documents or amendments.
- Version Control and Provenance: Each node and relationship will include timestamps, source references, and version identifiers for traceability.

3.4.5 Phase 5 – Backend Integration (FastAPI)

FastAPI services will provide access to the KG for other components:

- Graph Query APIs: RESTful and GraphQL endpoints to fetch statutes, relationships, and entity graphs.
- Integration with Reasoning Module: Enabling the reasoning engine to reference statutes, relationships, and precedent data dynamically during chain-of-thought generation.
- Monitoring and Logging: Capturing API usage, query times, and system health for maintenance and scaling.

3.4.6 Phase 6 – Front-End Features (React)

Though the primary consumer of the KG will be the reasoning module, a user-friendly interface will also be available for direct interaction:

- Entity Browser: Visual exploration of statutes, cases, and parcels through interactive graph visualizations.
- Search and Query Tools: Simple search for specific laws or complex query builders for advanced users.
- Graph Visualization: Libraries like D3.js or Cytoscape.js to render relationship maps that are intuitive and interpretable.

3.4.7 Phase 7 – Testing and Validation

To ensure accuracy and reliability, the KG will undergo rigorous testing:

Technical Testing

- Unit Tests for API endpoints, data pipelines, and entity linking scripts.
- Integration Tests to validate seamless interaction between KG, reasoning module, and front-end.

Expert Validation

- Collaboration with legal experts to evaluate:
 - Accuracy of entity linking.
 - Correctness of relationships and references.
 - Completeness of statutory coverage.

Performance Metrics

- Query latency and throughput.
- Accuracy scores for entity extraction and relationship mapping.
- Expert-rated usability of graph visualizations.
- LLM-AS-JUDGE method will be explored.

3.4.8 Phase 8 – Continuous Updates and Learning

Given the dynamic nature of legal data, the KG will support real-time or scheduled updates:

- Automated Data Feeds from gazettes, registries, and court databases.
- Incremental Updates to avoid downtime or performance degradation.

- Feedback Loop: Incorporating user corrections and expert feedback to refine entity mappings and ontology over time.

3.4.9 Deployment and Scalability

- Containerized Deployment: Using Docker to enable easy setup, scaling, and portability.
- Cloud Hosting: Hosting in a secure environment with robust backup and failover strategies.
- Scalability: Horizontal scaling of graph databases and APIs to support growing datasets and user traffic.

3.5 System Architecture

The overall architecture of the LegalVision platform follows a layered design, ensuring modularity, scalability, and clear separation of concerns between user interaction, application logic, data management, and external integrations.

A detailed system architecture diagram is provided in [Appendix-II](#). The platform is divided into four primary layers, described below:

3.5.1 Presentation Layer (React)

The front-end interface, built with React, enables:

- Submission of legal queries and property documents.
- Visualization of reasoning steps, chains, and logic graphs.
- Exploration of knowledge graph entities and relationships.
- A responsive and intuitive experience optimized for legal professionals and academic users.

3.5.2 Application Layer (FastAPI)

This layer serves as the core logic layer and orchestrates:

- **API Gateway:** Handles authentication, authorization, and request routing.
- **Reasoning Module:** Processes user queries, triggers prompt chaining and chain-of-thought reasoning, and integrates outputs from the LLM.
- **Knowledge Graph Module:** Executes entity and relationship queries to provide statute references, case connections, and contextual data to the reasoning engine.

3.5.3 Data Layer (MongoDB & Neo4j)

This layer manages data storage and retrieval:

- **MongoDB:** Stores user queries, reasoning chains, usage logs, and evaluation records.
- **Neo4j GraphDB:** Stores statutes, cases, entities, and relationships for knowledge graph queries.
- **Object Storage (S3 or Local):** Stores uploaded documents, OCR outputs, and processed files for traceability and future training use.

3.5.4 External Integration Layer

This layer connects the system to external resources:

- **Legal Data Sources:** Automated ingestion of gazettes, statutes, and case law for continuous updates.
- **LLM Providers:** Secure access to hosted LLM APIs (e.g., GPT-4) or self-hosted models for inference and reasoning tasks.

This layered approach ensures a scalable and maintainable architecture, allowing independent updates to the front-end, back-end, and data modules while maintaining robust integration for reasoning and knowledge graph functionalities.

3.6 Context Diagram

The context diagram illustrates the high-level interaction between the main components of the *Knowledge-Driven AI Framework for Legal Contract Understanding and Trust Assessment*. This framework integrates four core modules that interact seamlessly to analyze documents, reason through legal content, build dynamic knowledge graphs, and evaluate bias and risk.

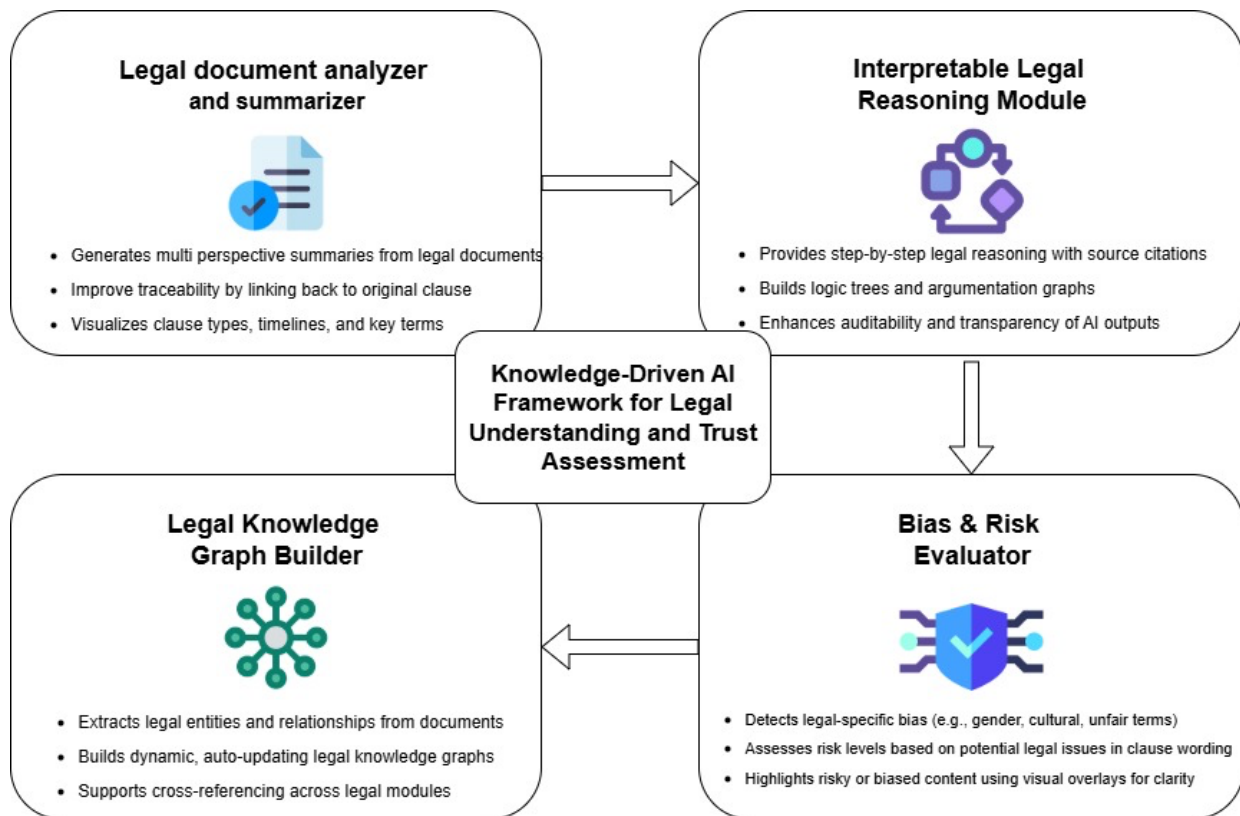


Figure 1: Context Diagram

3.7 Risk Management

Identifying potential risks and implementing effective mitigation strategies is critical to ensure the stability, reliability, and ethical compliance of the system. The table below summarizes the key risks, their severity, and the strategies adopted to minimize their impact.

Table 6: Risk Management

Risk	Severity	Mitigation Strategy
Incomplete or outdated legal data	High	Establish automated data ingestion pipelines from official sources (gazettes, registries, and case law portals) and schedule regular data audits.
Model bias or inaccurate reasoning outputs	High	Use diverse training datasets, integrate a bias detection module, and include legal expert reviews in the validation cycle.
Data privacy breaches	High	Anonymize sensitive data, enforce encryption for storage and transmission, and restrict system access with authentication and role-based permissions.
Integration failures between modules	Medium	Conduct thorough integration testing, implement error logging, and use rollback mechanisms to recover from failures.
Performance or scalability issues	Medium	Optimize API and database queries, use containerized deployments for easy scaling, and monitor system load to plan for upgrades.
User misinterpretation of outputs	Medium	Include clear disclaimers in the UI, provide reasoning transparency with citations, and encourage professional validation of results.
Project timeline delays	Low	Use Agile sprints with regular progress reviews and adjust schedules proactively based on milestone tracking.
Hardware or resource limitations	Low	Use cloud-based solutions for model hosting and backups to ensure high availability and recovery options.

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

The functional requirements define what the system must do to deliver its intended outcomes. They cover data ingestion, processing, visualization, and integration with other LegalVision components.

Table 7: Functional Requirements

ID	Requirement	Description
FR-01	Integrated Data Ingestion	Collect and preprocess statutes, gazettes, case law, cadastral records, and registry data from multiple sources into a unified, structured format.
FR-02	Machine-Readable Storage	Convert unstructured and static documents (PDFs, scans) into structured, queryable data suitable for graph representation.
FR-03	Ontology-Based Representation	Map legal entities and their relationships using a domain-specific ontology that supports interoperability and versioning.
FR-04	Graph Query and Search	Enable semantic and keyword-based queries, allowing users to retrieve statutes, relationships, and related legal contexts efficiently.
FR-05	Graph Visualization	Provide an interactive front-end for visualizing relationships between statutes, parcels, judgments, and administrative actions.
FR-06	Dynamic Updates	Continuously update the KG when new gazettes, amendments, or registry changes are published.
FR-07	API Access	Offer REST and GraphQL APIs to support integration with reasoning modules and external applications.
FR-08	Data Export	Allow exports in JSON, CSV, or PDF formats for legal professionals and policy analysts.

4.2 Non-Functional Requirements

The non-functional requirements define how the system should perform to ensure reliability, scalability, and usability in real-world legal workflows.

Table 8: Non-Functional Requirements

ID	Requirement	Description
NFR-01	Performance	Queries on the knowledge graph should return results within 3 seconds for standard search requests and within 5 seconds for complex relational queries.
NFR-02	Scalability	The system must scale horizontally to handle growing datasets, increasing to millions of nodes and edges without performance degradation.
NFR-03	Security	Implement role-based access controls, encryption (TLS/SSL), and secure authentication to comply with the Sri Lanka Personal Data Protection Act (No. 9 of 2022).
NFR-04	Reliability	Maintain a 99% uptime, supported by automated backups, failover servers, and real-time health monitoring.
NFR-05	Auditability and Traceability	All data changes, query logs, and updates should be recorded for compliance, debugging, and legal audits.
NFR-06	Usability	The interface should be intuitive and accessible to legal professionals with minimal technical training.
NFR-07	Maintainability	The architecture should be modular, allowing easy updates to APIs, ontologies, and visualization components without disrupting live services.
NFR-08	Interoperability	Ensure compatibility with external reasoning engines and legal analytics platforms through standards like RDF, OWL, and SPARQL.

4.3 Expected Test Cases

Table 9: Expected Test Cases

Test Case ID	Scenario	Expected Result
TC-01	Data Ingestion – Statutes Upload a new statute (PDF or scanned document) into the ingestion pipeline.	Text is extracted, cleaned, normalized, and stored in the graph with proper entity tagging and version metadata.
TC-02	Data Ingestion – Case Law Integrate a set of judgments into the KG.	Entities (e.g., parties, statutes referenced) and relationships (e.g., case-to-statute links) are correctly created and linked.
TC-03	Gazette Notifications Upload a new land acquisition notice from the Gazette.	Notification is processed and linked to the relevant parcels and statutes in the KG.
TC-04	Cadastral Data Integration Import parcel and survey records from Survey Department data.	Parcel nodes are created, linked to ownership and legal references, and spatial metadata is preserved.
TC-05	Graph Query – Basic Search Search for a specific statute by keyword.	Accurate retrieval of statute node, metadata, and linked relationships.
TC-06	Graph Query – Complex Relationship Query all judgments referencing a specific statute.	Graph returns a complete set of related cases, with correct relationships and source citations.

TC-07	Visualization View an interactive graph of a parcel, its owners, and linked statutes.	Clear, accurate visualization of nodes and edges, with drill-down options for detailed information.
TC-08	Dynamic Updates Add a new amendment or registry update.	KG is updated incrementally without downtime; changes are visible in queries and visualizations.
TC-09	API Query – REST/GraphQL Call API to retrieve relationships for a specific case.	Accurate JSON or GraphQL response, matching front-end query results, delivered within performance limits.
TC-10	Performance Under Load Run concurrent queries (100+ users).	System maintains response times within defined performance thresholds ($\leq 3s$ basic, $\leq 5s$ complex).
TC-11	Security & Access Control Attempt unauthorized access to restricted datasets.	Access is denied, with secure logging of the attempt and no data exposure.
TC-12	Audit and Logging Review logs after a data update and query session.	All actions are traceable, including user, timestamp, and change details, in compliance with audit standards.
TC-13	Backup and Recovery Simulate server failure and restore from backup.	System restores fully with no data loss and minimal downtime.
TC-14	Integration with Reasoning Module Run a reasoning engine query referencing multiple statutes.	Reasoning engine retrieves accurate, real-time data from the KG and generates consistent outputs.

COMMERCIALIZATION

The knowledge graph component offers strong commercialization opportunities in Sri Lanka's property law domain. Legal firms and notaries can use it to streamline deed verification and cross-referencing of statutes, while government registries and survey departments can adopt it for more reliable land administration. Financial institutions can integrate the system into loan approval workflows for property-backed securities, enhancing risk assessments.

By providing a queryable, continuously updated, and explainable platform, this component reduces errors, improves efficiency, and ensures transparency in legal decision-making. Its modular design also enables future expansion into other legal domains, supporting subscription-based services, licensing models, and integration with e-government initiatives.

BUDGET AND BUDGET JUSTIFICATION

Table 10: Budget and budget justification

Component	Cost (LKR)	Justification
Travel & Meetings with Law College Experts	10,000	To get legal counsel and obtain datasets. This includes transportation, refreshments, and coordination (around 2-3 trips).
Data Collection	5,000	Purchase or get legal datasets required for analysis and model training (e.g., property law case reports and registries).
Research Paper Publication	100,000	Includes the expense of publishing in recognized journals and conferences.
Internet Charges	24,000	Continuous internet connectivity (2,000 LKR x 12 months) is required for dataset downloads, literature surveys, virtual meetings, and cloud-based experiments.

Total Estimated Budget: 139,000 LKR

The proposed financing addresses the research's primary demands. Travel and meeting expenses make it easier to communicate with legal specialists and obtain datasets. Data collection ensures that trainers and analysts have access to reliable legal texts, such as property law records and case reports. Publication fees enable the dissemination of discoveries in recognized journals and conferences, ensuring academic credibility. Internet charges allow for ongoing access to dataset downloads, virtual collaboration, and cloud-based experimentation. Overall, funding is balanced, allowing for both good research and effective dissemination.

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APPENDICES

Appendix-I Resources for Sri Lankan Property Law

1. Official Legislation Portals

- SriLankaLaw Revised Statutes – Offers complete texts of key property laws such as the Registration of Documents Ordinance (Cap. 300), outlining definitions, registration procedures, and legal effects of deeds, leases, and mortgage instruments [LawGratis+2ResearchGate+2ResearchGate+4Sri Lanka Law+4FAOLEX+4](#).
- FAOlex / Government PDF – Provides the full text of the Land Development Ordinance (No. 19 of 1935), governing the development and alienation of state land [Wikipedia+4FAOLEX+4Land Portal+4](#).
- Wikipedia Summaries:
 - Land Development Ordinance – Summarizes history and provisions enabling state land resettlement [SearchWorks+9Wikipedia+9FAOLEX+9](#).
 - State Lands Ordinance (No. 8 of 1947) – Covers Crown land classification and governance mechanisms [Wikipedia+1](#).
- SriLankaLaw Articles – Example: Article on the Registration of Title Act, No. 21 of 1998, detailing its transformative role in property title clarification [Lanka Law+15Lanka Law+15resources.ials.sas.ac.uk+15](#).

2. Legal Research and Case Law

- vLex Sri Lanka Collection – A subscription-based platform offering a vast repository of Sri Lankan legal documents—consolidated Acts, Supreme Court Reports, New Law Reports, etc. [vLex](#).

- Bodleian Law Cases Guide – Lists open-access resources for Sri Lankan Supreme Court and Court of Appeal judgments (New Law Reports from 1878 onward) [libguides.bodleian.ox.ac.uk+1](https://libguides.bodleian.ox.ac.uk/1).
- GlobaLex (NYU Law) – An authoritative guide to Sri Lankan legal research, including colonial legal history, court system structures, and resource bibliographies [GlobaLex+2resources.ials.sas.ac.uk+2](https://globalex.ials.sas.ac.uk/2).

3. Legal Overviews & Context

- LawGratis “Property Law in Sri Lanka” – A blog entry (Mar 2025) that covers legal foundations such as Roman-Dutch and English Common Law influences, the Land Development Ordinance, Registration of Titles, and other property frameworks [LawGratis+1](#).
- LandPortal – Sri Lanka Narrative – Offers insights into land governance, including the LDO’s context and implications in settling state lands [Land Portal+1](#).

4. Government Gazette

- Sri Lanka Gazette – The official publication for statutory announcements, including enactment of new laws, land notifications, government proclamations, and legislative updates [Wikipedia](#).

5. Institutional & Library Resources

- Law & Society Trust (LST) – A premier Sri Lankan legal research institution offering access to Acts, Law Reports, Gazette archives, and legal periodicals (not directly cited here, but highly recommended).

- Survey Department & Ministry Resources – While not fully captured here, these entities publish land and cadastral data and may provide additional legal frameworks around land mapping and regulation (see Ministry of Urban Development context) [Wikipedia](#).

6. Additional Tools & Educational Resources

- Scribd Presentation – Provides educational summaries comparing the Registration of Documents Ordinance (1927) and the Registration of Title Act (1998), highlighting procedural evolution [ResearchGate+6Scribd+6Studocu+6](#).

Resource Summary Table

Table 11: Resource Summary Table

Resource / Platform	Description
SriLankaLaw Revised Statutes	Official texts for statutes like the Registration of Documents Act
FAO / Government PDFs	Access to full Land Development Ordinance texts
Wikipedia Summaries	Useful overviews of major land laws (with citations)
SriLankaLaw Articles	Insights into modern laws like title registration
vLex Sri Lanka	Paid platform with legislation and case law compilation
Bodleian Law Cases Guide	Free access guide to older Sri Lankan case law
GlobaLex (NYU Law)	Legal system overview and resource bibliography
LawGratis	Blog overview of property law frameworks
LandPortal Narrative	Governance context and land administration discussion
Sri Lanka Gazette	Primary source for new laws and notifications
Institutional Repositories (e.g., LST)	Deep archives of statutes, judgments, gazettes
Scribd Educational Materials	Visual comparisons of deed vs title systems

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Appendix-II System Architecture Diagram

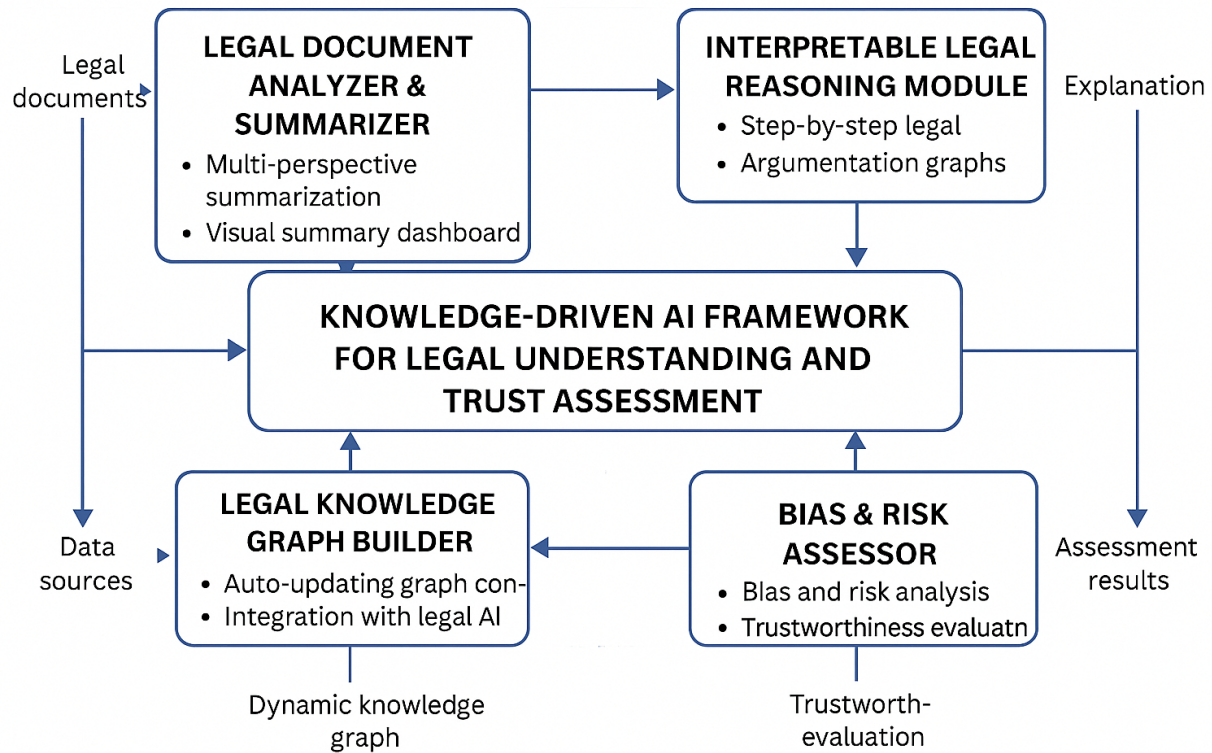


Figure 2: System Architecture Diagram

Appendix-III Gantt Chart

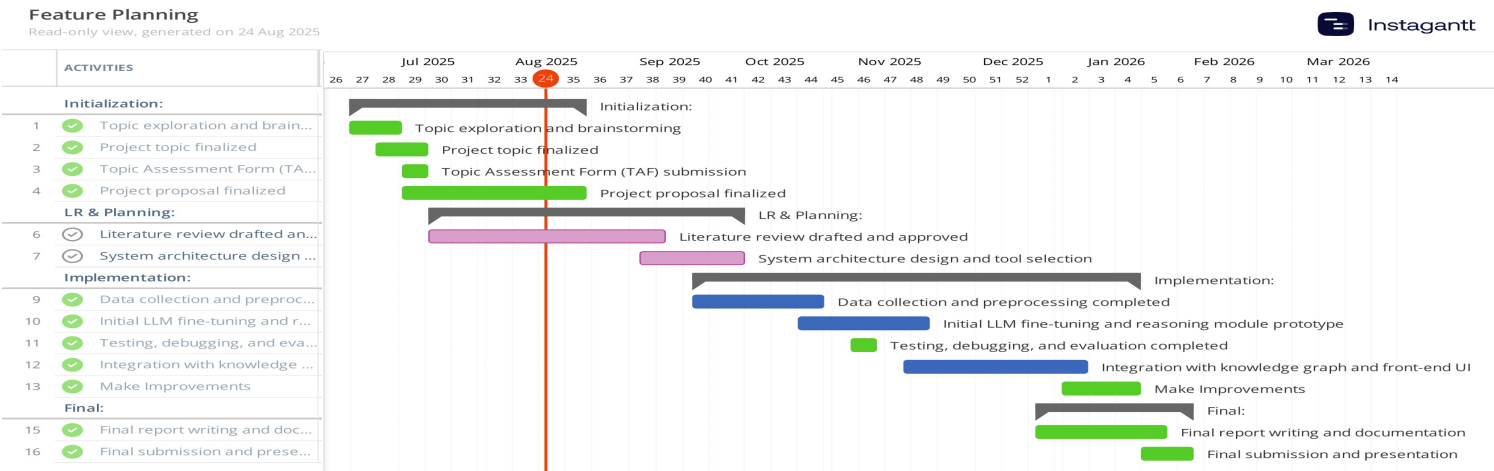


Figure 3: Gantt Chart